Complete degree and closure comparison

Fixed p37 seed0 diagnostic · 2026-10-25

Terminal exit 0 at 07:20:28 UTC. All 24 producers, twelve first audits, twelve A/A parities, the native reducer and final source/input/execution seals pass. Exactly the registered systems 0,1,2 from each phi/doubling arm are retained. Source/input freeze c30a3dc6068b precedes the managed 04:08:23 UTC launch. No collector, oracle or target solve was repeated.

Every native result

Arm / index	Method	Power	Working / product degree	Search Fp37 multiplications	Sparse-check multiplications
phi / 0	raw16	2	12 / 12	33,609,631	15,211
phi / 0	closure8	2	7 / 12	916,980	15,983
phi / 1	raw16	none	16 / —	142,494,025	0
phi / 1	closure8	none	8 / —	569,173	0
phi / 2	raw16	1	12 / 12	33,569,618	14,675
phi / 2	closure8	1	7 / 12	926,804	16,686
double / 0	raw16	1	12 / 12	33,603,044	14,614
double / 0	closure8	1	7 / 12	949,579	16,857
double / 1	raw16	none	16 / —	142,494,025	0
double / 1	closure8	none	8 / —	569,173	0
double / 2	raw16	1	12 / 12	33,716,337	15,142
double / 2	closure8	1	7 / 12	956,093	17,595

All rows are verified-first-aa. Inversion exponentiation products are included in multiplication counts. The native table retains additions, inversion calls and every input/candidate/reference/audit/parity hash. Zero check-count means no certificate was produced; it is not a cost-free rejection. All six raw16 results exactly reproduce prior raw degrees 4..8, including ranks and counters. That requirement does not apply to closure8.

Working degree and original degree are different

Systems 0 and 2 produce independently checked certificates in both arms and both methods. Closure finds them at working degree 7, but their actual original-product degree is 12. Sparse verification is slightly more costly for its different multipliers and remains counted. No degree-7 original certificate or equality of certificate bytes is claimed.

Both index-1 systems find no certificate for powers 1,2,3 within the bounds. Raw16 raises the original Macaulay bound through 16. Closure8 caps working degree at 8 and tracks original products through 16. These are different bounded searches; closure failure does not exclude every original-degree-16 representation, and neither failure proves nonmembership at every degree or power. An explicit q-nonzero common-root witness or separate higher-power control is a future discriminator, not a result here.

Counts and attribution

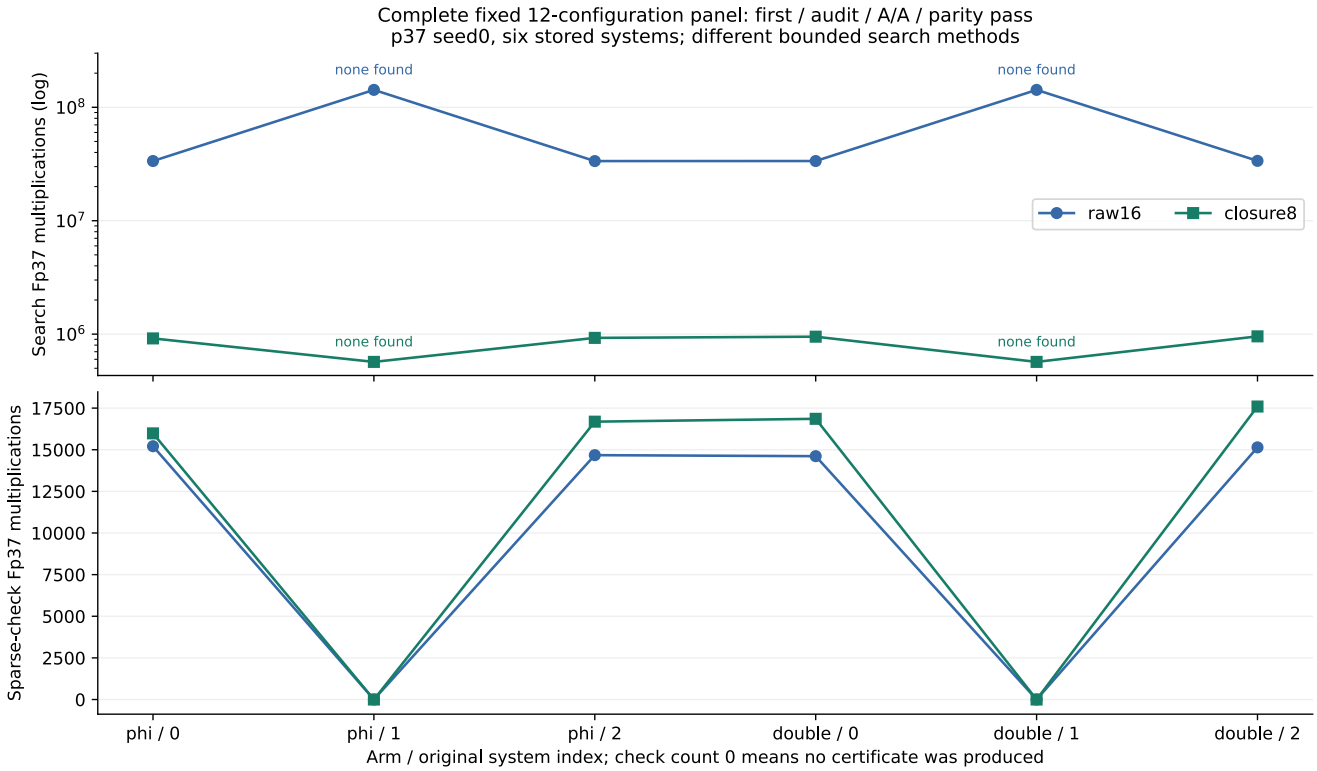


Figure 1: All twelve configurations; literal recorded native counts. Both methods pass their first/audit/A/A/parity checks. Negative labels retain the full registered search bounds; they do not mean universal nonexistence.

The construction-count reduction occurs for generic doubling as well as phi. This diagnostic does not establish a twist-specific construction advantage. Within each arm/index, both methods consume identical equations; the two maps produce different residual systems. The six systems are from one synthetic p37 seed0 fixture and are not independent secret samples.

The old phi system-0 F4 reference records 125,327 row-reduction multiplications for finding all roots. Even closure's 916,980 search products exceed that partial count. Finding all roots and constructing a rejection certificate are different tasks. No actual fallback/filter is executed here. These counts do not price root extraction, the remaining F4 work, construction, memory or the full pipeline.

The atlas already distinguishes raw Macaulay spans from degree-fall closure in another setting. No novelty, global atlas saturation, fastest-tool or asymptotic claim follows. Shared-host clocks remain unadmitted diagnostics. All complete-price fields are null.

Verification, scope and reproduction

The producer receives equation-only input and peeled abscissa. Prior roots, physical outcomes and old costs belong to the auditor. Successful certificates undergo independent sparse expansion against the original equations and checks of retained roots. The generic conditional rejection theorem is separately checked. The concrete phi-0 integer-lift proof has two retained memory failures and a distinct v3 attempt; its outcome is not inferred here. Full parser, field/group/decoder bridges remain open.

Native v2 gate passes five tests, including the prelaunch cap-versus-actual- degree regression. V1′s four-test source is preserved. Launcher controls refuse nested outer busy (64) and an existing phase command (65) without science output. The old v1 queued control was quiesced before admission. No terminal or interrupted producer is silently retried.

Each phase keeps one worker, common reservation, 8 GiB, 1,200-second outer and 1,180-second cooperative producer limits. Queue wait is separate. The original overflowing layout, corrected layout, native gates and partial historical progress snapshot remain unchanged in the versioned archive.

Requirement	Actual status
Twelve-configuration fixed comparison	Complete, all first/audit/A/A/parity and terminal seals pass.
Successful finite certificates	Sparse identity and retained-root checks pass.
Arbitrary-degree/power nonmembership for failures	Not established by bounded searches.
New unique relations, matrix rank or recovery	Not collected by this diagnostic.
Complete cost attributable to twist	Open. Counted arithmetic excludes other work, memory, I/O, calibration and matched strongest IC/rho/external construction controls.
Original 189 + 21	Complete and not repeated; no consistent twist-specific full cold gain.
85 target / 42 certificate panels	Original managed owners, inputs and full workloads retained.

The separate all-42 finite-residual census passes both processes and exact parity, retaining physical sets and independent ranks. Its 300 phi versus 150 doubling eligible systems are a counterfactual skip count; no filter is deployed. It is not combined with this construction diagnostic to invent a measured pipeline result. Multisets require their own replay and scope.

Study: zero_fiber_collector/degree_closure_probe/. See PROTOCOL.md, FREEZE.json, native sources, run-panel.sh, run-phase.sh and workload-v1. Raw final table: parent runs/degree-closure-probe-panel-v1/analysis/ summary.json. Compact final receipts are under results/degree-closure-probe-v1/final/. Extract the complete versioned raw archive under the parent collector; README.md gives checksum and reproduction commands. The prior reference corpus remains a prerequisite.

Workload SHA-256: `254581dfebfcb992e8ce0c6b7896cbd7cb402259b0a5b72e34fd8c711cc0c88a` Runner SHA-256: `d6e4f2f6d562f0a67bd5735123d63a6d744c9e685939f4c73d9c427ad0c7a92b` The prelaunch control figure/PDF remain historical. Every actual final PDF page is inspected; unlisted benchmark copies preserve its bytes and the SVG/150 dpi PNG. Canonical IC/rho graphs, leaderboard and benchmark index are unchanged because full-price, reference and VALIDATION gates remain unmet.